

Homework 4

(due May 4 @15:55, in Blackboard)

1) Let f be a differentiable function with $\text{dom } f = \mathbf{R}^n$.

a) If ∇f is Lipschitz continuous with parameter $L > 0$, show that $g(x) := \frac{L}{2}\|x\|_2^2 - f(x)$ is convex. In addition, if x^* is a minimizer of f show that

$$\frac{1}{2L}\|\nabla f(x)\|_2^2 \leq f(x) - p^* \leq \frac{L}{2}\|x - x^*\|_2^2, \quad \forall x$$

b) If f is strongly convex with parameter $m > 0$ show that if there is a minimizer x^* , it is unique and it holds that

$$\frac{m}{2}\|x - x^*\|_2^2 \leq f(x) - p^* \leq \frac{1}{2m}\|\nabla f(x)\|_2^2, \quad \forall x$$

c) Explain the implications of a), b) for optimization algorithms.

d) If f satisfies the assumptions of both a) and b) show that

$$(\nabla f(y) - \nabla f(x))^T (y - x) \geq \frac{mL}{m+L}\|y - x\|_2^2 + \frac{1}{m+L}\|\nabla f(y) - \nabla f(x)\|_2^2, \quad \forall x, y$$

[Hint: The function $g := f - \frac{m}{2}\|\cdot\|_2^2$ is convex and ∇g is Lipschitz continuous with parameter $L - m$. This implies that $(\nabla g(y) - \nabla g(x))^T (y - x) \geq \frac{1}{L-m}\|\nabla g(y) - \nabla g(x)\|_2^2, \forall x, y]$

2) Analyze the gradient descent method for each of the following cases.

a) f has L -Lipschitz gradient. Show that

$$f(x^{(k)}) - f(x^{(k-1)}) \leq -t^{(k)} \left(1 - \frac{Lt^{(k)}}{2}\right) \|\nabla f(x^{(k-1)})\|_2^2,$$

therefore for $t^{(k)} \in (0, \frac{2}{L})$ GD is a descent method.

[Hint: Use the quadratic upper bound (Hw1, problem 6a)]

b) In addition to the assumption in a), f is lower-bounded. For fixed step-size $t = \frac{1}{L}$, show that:

$$\min_{i=0,\dots,k} \|\nabla f(x^{(i)})\|_2^2 \leq \frac{2L}{k+1} (f(x^{(0)}) - p^*)$$

c) In addition to the assumptions in b), f is convex and attains the minimum at x^* . Show that:

$$f(x^{(k)}) - p^* \leq \frac{L}{2k} \|x^{(0)} - x^*\|_2^2$$

d) In addition to the assumptions in c) f is strongly convex with parameter $m > 0$. Show that for fixed step-size $t \in (0, \frac{2}{m+L}]$ it holds that:

$$\|x^{(k)} - x^*\|_2^2 \leq \rho^k \|x^{(0)} - x^*\|_2^2; \quad \rho := \left(1 - t \frac{2mL}{m+L}\right)$$

[Hint: Use the bound from problem 1d]

For the value of t that minimizes the upper bound, express ρ as a function of the condition number $\kappa := \frac{L}{m}$, and also derive a bound for $f(x^{(k)}) - p^*$ as a function of $f(x^{(0)}) - p^*$.

e) Explain the implications of parts b) (for non-convex optimization) and parts c),d) for convex optimization by analyzing the number of iterations needed to attain a given accuracy ϵ .

3) True or False? Briefly explain the reason.

- In descent methods, the particular choice of search direction does not matter so much.
- In descent methods, the particular choice of line search does not matter so much.
- When the gradient method is started from a point near the solution, it will converge very quickly.
- When Newton's method is started from a point near the solution, it will converge very quickly.
- Newton's method with step size $t^{(k)} \equiv 1$ always works; damping (i.e., using $t^{(k)} < 1$) is used only to improve the speed of convergence.
- Using the gradient method to minimize $f(Ty)$, where $Ty = x$ and T is non-singular, can greatly improve the convergence speed when T is chosen appropriately.
- Using Newton's method to minimize $f(Ty)$, where $Ty = x$ and T is non-singular, can greatly improve the convergence speed when T is chosen appropriately.

4) Develop code for gradient descent and Newton's method and use it to solve the following Machine Learning problems. You may use real datasets from the UCI Machine Learning Repository.

- Linear regression:* Given N examples $\{(a_i, b_i)\}_{i=1}^N$ ($a_i \in \mathbf{R}^d, b_i \in \mathbf{R}$), the goal is to build a linear prediction model so as to

$$\text{minimize } f(x) := \sum_{i=1}^N \|a_i^T x - b_i\|_2^2$$

- Logistic regression:* Given N examples $\{(a_i, y_i)\}_{i=1}^N$ ($a_i \in \mathbf{R}^{d+1}$ is the feature and $y_i \in \{0, 1\}$ is the corresponding label), the goal is to build a binary classifier so that $\text{sign}(a_i^T x)$ predicts the label y_i . This can be accomplished by aiming to

$$\text{minimize } f(x) := \sum_{i=1}^N \left\{ \log \left(1 + e^{a_i^T x} \right) - (1 - y_i) a_i^T x \right\}$$

[Given a feature $\bar{a}_i \in \mathbf{R}^d$ in UCI, set $a_i := (\bar{a}_i, 1)$ (augmented feature); this is done so that $\mathbf{R}^d$ is separated in two halfspaces, each corresponding to a class]

- For each problem analyze the properties of the objective (gradient Lipschitz continuity and strong convexity). Estimate the values L, m from the chosen datasets. [For any ill-conditioned instances you may augment the objective with the term $\frac{1}{2}\gamma\|x\|_2^2, \gamma > 0$; explain how you chose γ]
- For each problem, use CVXPY to compute the optimal solution and value (x^*, p^*) .
- For each problem and both methods generate two figures, by plotting i) $\|x^{(k)} - x^*\|_2$ and ii) $f(x^{(k)}) - p^*$ vs. iteration count k . [Use logarithmic scale]
- For each problem, report the run-time required for each method to reach an RMSE (rooted-mean squared error) $\frac{1}{\sqrt{d}}\|x^{(k)} - x^*\|_2 \leq \epsilon$, for $\epsilon \in \{10^{-2}, 10^{-3}, 10^{-4}\}$.
- Explain your results in view of the theoretical analysis.

5) Problem 10.15(a) from the reference book.

6) Develop code for solving an LP in the following standard form using the barrier method.

$$\begin{aligned} &\text{minimize} && c^T x \\ &\text{subject to} && Ax = b, x \succeq 0 \end{aligned}$$

- Illustrate the progress of the algorithm in a properly chosen random instance ($n = 100, m = 500$, where $A \in \mathbf{R}^{n \times m}$); additionally, compare with CVXPY and interpret your findings.
- For fixed n , solve multiple instances by varying $m \in [10, 1000]$ to report the number of Newton steps needed. Compare with the expected relation (11.29) in the reference book and interpret your results.